

<thread> Header in C++ - Namespaces and Components

Namespace Diagram

```
<std::thread>
|
|-- std          <- Main namespace
|  |-- thread    <- Class to manage threads
|   |-- id       <- Type for thread identifiers
|   |-- `-- hardware_concurrency() <- Static function
|  |-- `-- this_thread <- Sub-namespace
|     |-- get_id()    <- Gets ID of current thread
|     |-- yield()     <- Yields execution
|     |-- sleep_for() <- Sleep for duration
|     |-- `-- sleep_until() <- Sleep until time point
```

Component Descriptions

std::thread

Class to create and manage threads.

std::thread::id

A unique identifier for a thread.

std::thread::hardware_concurrency()

Returns the number of concurrent threads supported by hardware.

std::this_thread::get_id()

Returns the ID of the current thread.

std::this_thread::sleep_for()

Sleeps the current thread for a given duration.

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`std::this_thread::sleep_until()`

Sleeps the current thread until a specified time point.

`std::this_thread::yield()`

Suggests to the implementation to reschedule execution to allow other threads to run.